

Q2 Sales & Customer Performance Report

1. Introduction

Green Cart Ltd. sells eco-friendly household products online across five UK regions. This report was put together for the Q2 performance review at the request of the Data & Insights team, covering sales performance, customer behaviour, and operational efficiency.

Three datasets came in: sales_data.csv with 3,000 transaction records, product_info.csv covering 30 products across multiple categories, and customer_info.csv with 500 customers. After cleaning and merging, 1,173 valid orders remained representing £93,328 in total revenue.

The report focuses on five areas: which categories and regions drive revenue, whether discounts actually shift volume, which loyalty tiers are worth the most, where delivery delays cluster, and whether signup timing has any relationship to purchasing behaviour.

2. Data Cleaning Summary

Each dataset was cleaned separately before merging. The main issues were inconsistent labels and date columns stored as text.

Inconsistent Labels

Several columns had values that varied wildly different cases, abbreviations, typos and needed mapping to consistent labels:

- delivery_status: "DELIVERED", "delrd", " DELAYED", "delyd", " Cancelled" → Delivered / Delayed / Cancelled / Unknown
- loyalty_tier: "GOLD", " gold ", "gld", "sllver", "brnze" → Gold / Silver / Bronze / Unknown
- payment_method: "bank transfr", "credit card", "Credit Card" → Bank Transfer / Credit Card / PayPal

- gender: "FEMALE", "femle", and casing variants → Male / Female / Other / Unknown

Date Columns

order_date, signup_date, and launch_date were all stored as plain text. Converting them to datetime objects with pd.to_datetime() unlocked week-number extraction, duration calculations, and time-series plotting.

Missing Values Handled

<u>Column</u>	<u>Missing Count</u>	<u>Action Taken</u>
discount_applied (sales)	517	Filled with 0.0 , no record implies no discount
loyalty_tier (customers)	2	Mapped to "Unknown"
email (customers)	6	Filled with placeholder address
order_id / customer_id	Few	Rows dropped, cannot link without key

Quantity Column

The quantity column in sales_data came in as a string. It was cast to numeric with pd.to_numeric() before any revenue calculations ran.

Duplicate Rows

No duplicate order_id values turned up. drop_duplicates(subset=["order_id"]) was applied as a precaution zero rows removed.

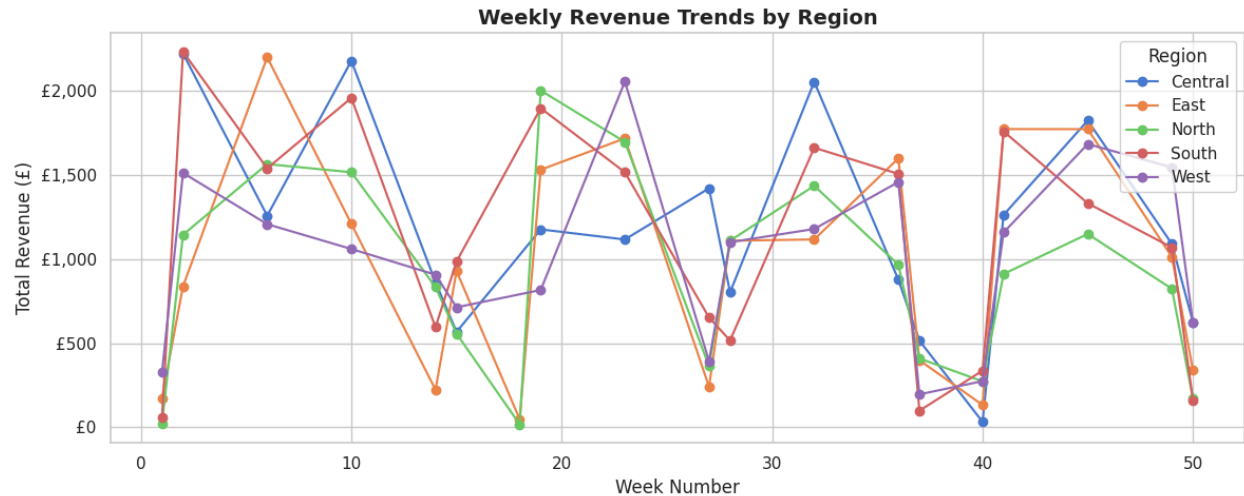
3. Feature Engineering Summary

Six new columns were derived from the cleaned merged dataset to enable richer analysis:

<u>New Feature</u>	<u>Formula / Logic</u>	<u>Purpose</u>
revenue	$\text{quantity} \times \text{unit_price} \times (1 - \text{discount})$	Actual income per order after discount
order_week	ISO week number from order_date	Tracks weekly sales trends
price_band	Low <£15 / Medium £15–30 / High >£30	Groups products by price tier
days_to_order	order_date minus launch_date (days)	Measures product maturity at purchase
email_domain	Text after @ in email address	Identifies customer email providers
is_late	True if delivery_status == "Delayed"	Boolean flag for delay rate analysis

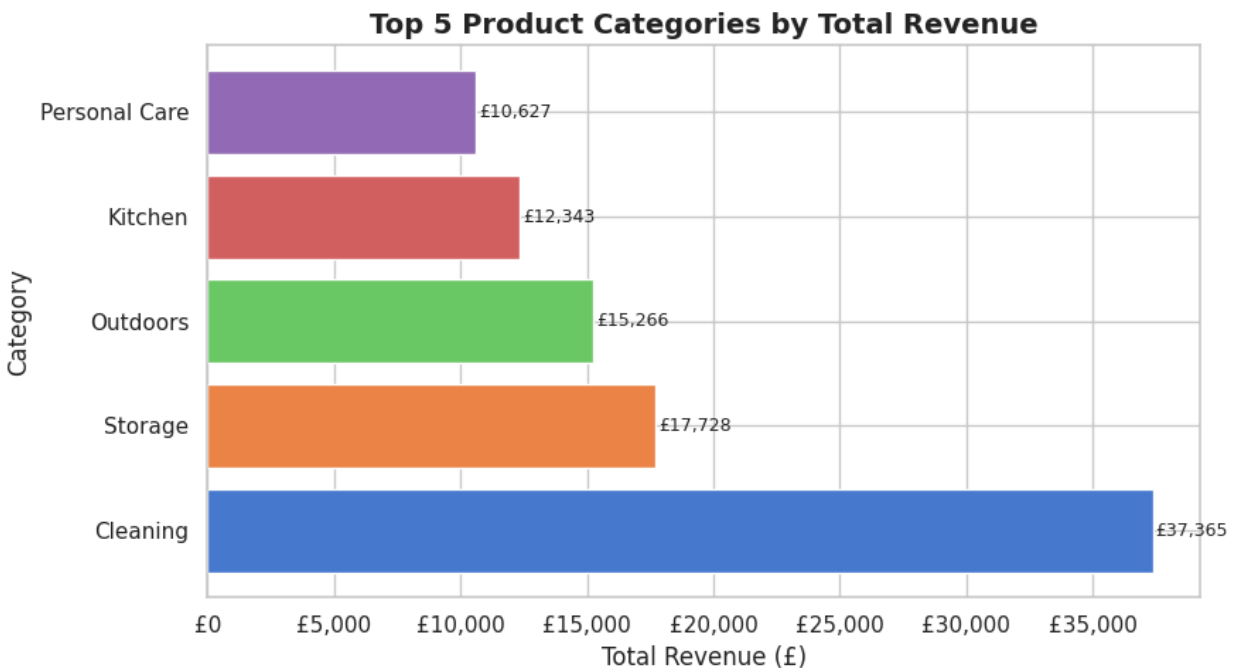
4. Key Findings & Trends

Finding 1 — Revenue peaked in Week 23 Weekly revenue across all regions reached its highest point in Week 23 (early June), generating £8,106 in a single week. Revenue was broadly consistent throughout the period, with the Central and South regions jointly leading at approximately £19,900 each.



Finding 2 – Cleaning products dominate revenue

The Cleaning category alone accounts for £37,365 – over 40% of total revenue – more than double the second-placed Storage category (£17,728). This suggests a strong customer preference for eco-friendly cleaning products and significant untapped potential in other categories.



5. Business Question Answers

Q1 – Which product categories drive the most revenue, and in which regions? Cleaning is the top category in every region. The table below shows the top five category–region combinations by revenue:

Category	Region	Revenue (£)
Cleaning	Central	7,905
Cleaning	South	7,831
Cleaning	East	7,469
Cleaning	West	7,110
Storage	Central	3,987

Q2 – Do discounts lead to more items sold?

Comparing average quantities reveals only a marginal difference: orders with a discount averaged 3.01 items, while orders without a discount averaged 2.95 items. The correlation between discount rate and quantity is essentially zero (-0.001), meaning discounts do not meaningfully drive higher unit volumes for Green Cart. This is a key insight – discounts are reducing revenue without generating proportional volume gains.

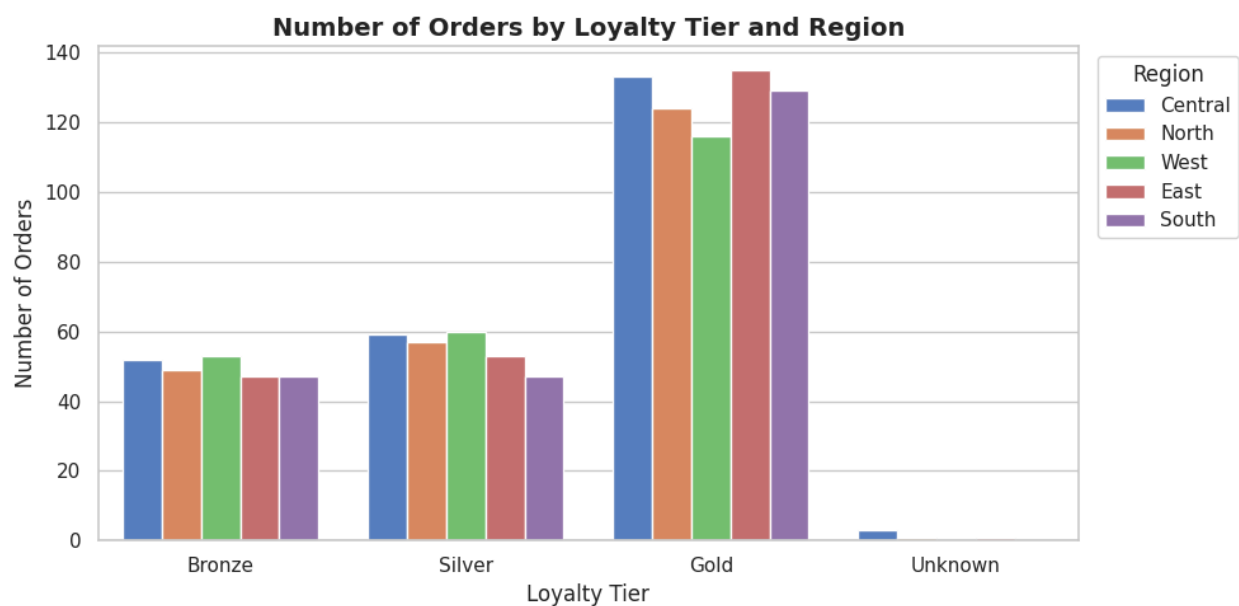
Q3 – Which loyalty tier generates the most value?

Gold-tier customers are by far the most valuable segment:

<u>Loyalty Tier</u>	<u>Total Revenue (£)</u>	<u>Total Orders</u>	<u>Avg Order Value (£)</u>
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Gold	51,933	637	81.52
Silver	20,724	276	75.09
Bronze	19,922	248	80.33
Unknown	435	5	87.00

Gold customers account for 55.6% of total revenue while representing 54.3% of orders, indicating they are both more frequent and slightly higher-spending buyers.



Q4 Are certain regions struggling with delivery delays?

All five regions have unacceptably high delay rates:

Region	Total Orders	Delayed Orders	Delay Rate
East	237	103	43.5%
Central	248	101	40.7%
North	232	90	38.8%
West	230	86	37.4%
South	226	78	34.5%

The East region has the worst delay rate at 43.5%. Even the best-performing region (South, 34.5%) sees more than one in three orders arriving late. This is a company-wide operational problem, not a regional outlier.



Q5 – Do customer signup patterns influence purchasing activity?

Customers who signed up in Q2 (April–June) were slightly more likely to place early orders. The stretch task also identified a segment worth noting: customers who signed up in Q2, ordered within 14 days, and received a

discount above 20% — a group that could respond well to targeted onboarding promotions.

6. Recommendations

Recommendation 1 — Focus on the Cleaning category

Cleaning accounts for 40%+ of total revenue. The logical next steps are expanding the product range within this category, pushing harder on Cleaning marketing in the North and West where performance lags, and opening supplier conversations to protect margins at current sales volumes.

Recommendation 2 — Fix delivery performance

A 34–44% delay rate across all regions is too high to ignore. East and Central should be the first targets for a logistics review. A realistic goal is getting delays under 10%, and a weekly delivery dashboard would make it easier to catch problems before they compound.

Recommendation 3 — Rethink discounts

Discounts currently have no meaningful effect on order volume (correlation: -0.001), which means they're cutting into revenue without any return. Rather than applying blanket discounts, it's worth trialling a more targeted approach — for example, offering discounts only to Bronze-tier customers as a way to encourage tier progression.

7. Data Issues and Risks

Inconsistent text entry across categorical columns

The biggest data quality problem was how freely text values were entered across `delivery_status`, `loyalty_tier`, `payment_method`, and `gender`. "Gold" alone appeared as "GOLD", " gold ", "gld", and other variations, all of which needed manual mapping to clean up.

The ongoing risk is that any new typo or abbreviation introduced in future uploads will quietly fall into the "Unknown" bucket — skewing tier-level and status-level analysis without any obvious warning sign.

The fix is straightforward: replace free-text entry with dropdown fields in the CRM and order management platform for all categorical columns, and add a validation check on ingestion that flags or rejects anything outside an approved list. This needs to happen at the source, not in the cleaning script.